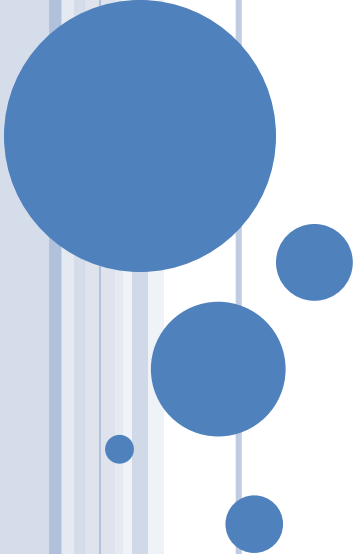




Relational Database Design Algorithms

DESIGNING A SET OF RELATIONS

Bottom-up Design Approach:

- Find all FD's from the mini-world.
- Constructs a minimal set of FDs
- Construct a Universal Relation Schema R
- Decompose R into a target set of 3NF or BCNF relations.
- Additional Criteria
 - **Attribute preservation**
 - **Dependency Preservation**
 - **Lossless (Non-additive) Join**



TESTING BINARY DECOMPOSITIONS FOR LOSSLESS JOIN PROPERTY:

- **Binary Decomposition:** decomposition of a relation R into two relations.
- **Lossless join test for binary decompositions:**
A decomposition $D = \{R_1, R_2\}$ of R has the lossless join property with respect to a set of functional dependencies F on R *if and only if* either
 - The FD $((R_1 \cap R_2) \rightarrow (R_1 - R_2))$ is in F^+ , or
 - The FD $((R_1 \cap R_2) \rightarrow (R_2 - R_1))$ is in F^+ .



ALGORITHM : FOR TESTING LOSSLESS JOIN PROPERTY

Input: A universal relation R ,
a decomposition $D = \{R_1, R_2, \dots, R_m\}$ of R , and
a set F of functional dependencies.

(b)

EMP		PROJECT			WORKS_ON		
SSN	ENAME	PNUMBER	PNAME	PLOCATION	SSN	PNUMBER	HOURS



PROPERTIES OF RELATIONAL DECOMPOSITIONS (8)

(c) $R = \{SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS\}$ $D = \{R_1, R_2, R_3\}$

$R_1 = EMP = \{SSN, ENAME\}$

$R_2 = PROJ = \{PNUMBER, PNAME, PLOCATION\}$

$R_3 = WORKS_ON = \{SSN, PNUMBER, HOURS\}$

$F = \{SSN \rightarrow \{ENAME\}; PNUMBER \rightarrow \{PNAME, PLOCATION\}; \{SSN, PNUMBER\} \rightarrow HOURS\}$

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1						
R_2						
R_3						

(original matrix S at start of algorithm)



PROPERTIES OF RELATIONAL DECOMPOSITIONS (8)

- (c) $R = \{SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS\}$ $D = \{R_1, R_2, R_3\}$
 $R_1 = EMP = \{SSN, ENAME\}$
 $R_2 = PROJ = \{PNUMBER, PNAME, PLOCATION\}$
 $R_3 = WORKS_ON = \{SSN, PNUMBER, HOURS\}$

$F = \{SSN \rightarrow \{ENAME; PNUMBER \rightarrow \{PNAME, PLOCATION\} ; \{SSN, PNUMBER\} \rightarrow HOURS\}$

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	a_1	a_2	b_{13}	b_{14}	b_{15}	b_{16}
R_2	b_{21}	b_{22}	a_3	a_4	a_5	b_{26}
R_3	a_1	b_{32}	a_3	b_{34}	b_{35}	a_6

(original matrix S at start of algorithm)



PROPERTIES OF RELATIONAL DECOMPOSITIONS (8)

- (c) $R = \{SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS\}$ $D = \{R_1, R_2, R_3\}$
 $R_1 = EMP = \{SSN, ENAME\}$
 $R_2 = PROJ = \{PNUMBER, PNAME, PLOCATION\}$
 $R_3 = WORKS_ON = \{SSN, PNUMBER, HOURS\}$

$F = \{SSN \rightarrow \{ENAME, PNUMBER\}; \{PNUMBER\} \rightarrow \{PNAME, PLOCATION\}; \{SSN, PNUMBER\} \rightarrow HOURS\}$

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	a_1	a_2	b_{13}	b_{14}	b_{15}	b_{16}
R_2	b_{21}	b_{22}	a_3	a_4	a_5	b_{26}
R_3	a_1	b_{32} a_2	a_3	b_{34} a_4	b_{35} a_5	a_6

(matrix S after applying the first two functional dependencies -
 last row is all "a" symbols, so we stop)



PROPERTIES OF RELATIONAL DECOMPOSITIONS (8)

LOSSLESS (NONADDITIVE) JOIN TEST FOR N -ARY DECOMPOSITIONS.

(A) CASE 1: DECOMPOSITION OF EMP_PROJ INTO EMP_PROJ1 AND EMP_LOCS

- (a) $R = \{SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS\}$ $D = \{R_1, R_2\}$
 $R_1 = \text{EMP_LOCS} = \{ENAME, PLOCATION\}$
 $R_2 = \text{EMP_PROJ1} = \{SSN, PNUMBER, HOURS, PNAME, PLOCATION\}$
 $F = \{SSN \rightarrow ENAME; PNUMBER \rightarrow \{PNAME, PLOCATION\}; \{SSN, PNUMBER\} \rightarrow HOURS\}$

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1						
R_2						



PROPERTIES OF RELATIONAL DECOMPOSITIONS (8)

LOSSLESS (NONADDITIVE) JOIN TEST FOR N -ARY DECOMPOSITIONS.

(A) CASE 1: DECOMPOSITION OF EMP_PROJ INTO EMP_PROJ1 AND EMP_LOCS

- (a) $R = \{\text{SSN}, \text{ENAME}, \text{PNUMBER}, \text{PNAME}, \text{PLOCATION}, \text{HOURS}\}$ $D = \{R_1, R_2\}$
 $R_1 = \text{EMP_LOCS} = \{\text{ENAME}, \text{PLOCATION}\}$
 $R_2 = \text{EMP_PROJ1} = \{\text{SSN}, \text{PNUMBER}, \text{HOURS}, \text{PNAME}, \text{PLOCATION}\}$

$F = \{\text{SSN} \rightarrow \text{ENAME}; \text{PNUMBER} \rightarrow \{\text{PNAME}, \text{PLOCATION}\}; \{\text{SSN}, \text{PNUMBER}\} \rightarrow \text{HOURS}\}$

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	b_{11}	a_2	b_{13}	b_{14}	a_5	b_{16}
R_2	a_1	b_{22}	a_3	a_4	a_5	a_6

(no changes to matrix after applying functional dependencies)



ALGORITHM: DECOMPOSITION INTO 3NF WITH DEPENDENCY PRESERVATION & LOSSLESS JOIN

Input: A universal relation R and a set of functional dependencies F on the attributes of R .

1. Find a minimal cover G for F
2. For each X of a FD in G , create a relation in D with attributes $\{X \cup \{A_1\} \cup \{A_2\} \dots \cup \{A_k\}\}$,
where $X \rightarrow A_1, X \rightarrow A_2, \dots, X \rightarrow A_k$ are the only FD's in G with X as LHS (X is the *key* of this relation).



EXAMPLE

- Consider $U(\text{ssn}, \text{pno}, \text{sal}, \text{phone}, \text{dno}, \text{pname}, \text{ploc})$
- Consider following FDs
 - FD1: $\text{ssn} \rightarrow \text{sal}, \text{phone}, \text{dno}$
 - FD2: $\text{pno} \rightarrow \text{pname}, \text{ploc}$
 - FD3: $\{\text{ssn}, \text{pno}\} \rightarrow \text{sal}, \text{phone}, \text{dno}, \text{pname}, \text{ploc}$
- Key: $\{\text{ssn}, \text{pno}\}$
- Step1: minimal cover
 - $G: \{\text{ssn} \rightarrow \text{sal}, \text{phone}, \text{dno}; \text{pno} \rightarrow \text{pname}, \text{ploc}\}$
- Step2:
 - $R1(\text{ssn}, \text{sal}, \text{phone}, \text{dno})$
 - $R2(\text{pno}, \text{pname}, \text{ploc})$
- Step3:
 - **$R3(\text{ssn}, \text{pno})$**



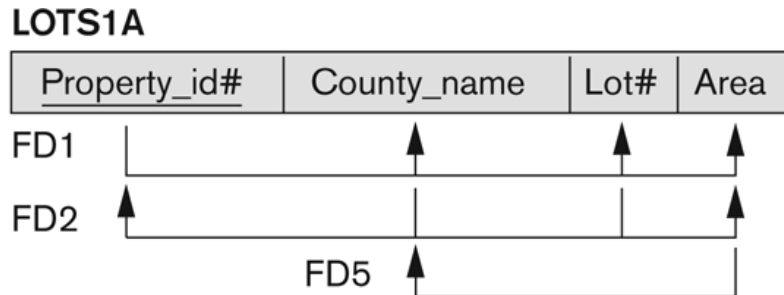
ALGORITHM: DECOMPOSITION INTO 3NF WITH DEPENDENCY PRESERVATION & LOSSLESS JOIN

Input: A universal relation R and a set of functional dependencies F on the attributes of R .

1. Find a minimal cover G for F
2. For each X of a FD in G , create a relation in D with attributes $\{X \cup \{A_1\} \cup \{A_2\} \dots \cup \{A_k\}\}$,
3. *If none of the relation in D contains a key of R , then create one more relation in D that contains key of R .*
4. Eliminate Redundant relations from D



EXAMPLE: PROPERTY LOT



- Consider LOTS1A is a Universal Relation with FD's
 - F: {P->LCA, LC->AP, A->C}
- Step1: Minimal cover
 - GX:{P->LC, LC->AP,A->C}
- Step2:
 - Design X: R1(P,L,C), R2(L,C,A,P), R3(A,C)
- Step4: Remove redundant relations
 - Design X: R2(L,C,A,P)



EXERCISE PROBLEM

- Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies
- $F = \{ \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$.
- What is the key for R ?
- Decompose R into 3NF relations
- Apply the synthesis algorithm to decompose R into 3NF relations
- Decompose R into BCNF relations.



SOLUTION

- AB is a key.
 - This can be determined by calculating AB^+ with respect to the set F.
- 2NF
 - R1 (A, B, C)
 - R2 (A,D,E,I,J)
 - R3 (B, F, G, H)
- 3NF
 - R1 (A, B, C)
 - R2.1 (A,D,E)
 - R2.2 (D, I, J)
 - R3.1 (B, F)
 - R3.2 (F, G, H)



DECOMPOSITIONS OF PREVIOUS QUESTION

- Determine whether each decomposition has
 - (1) the dependency preservation property, and
 - (2) the lossless join property, with respect to F .
- Also determine the normal form of each relation in the decomposition
- $R = \{A, B, C, D, E, F, G, H, I, J\}$
- $F = \{ \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$.

- $D1 = \{R1, R2, R3, R4, R5\}$;
 - $R1 = \{A, B, C\}$, $R2 = \{A, D, E\}$, $R3 = \{B, F\}$, $R4 = \{F, G, H\}$, $R5 = \{D, I, J\}$
- $D2 = \{R1, R2, R3\}$;
 - $R1 = \{A, B, C, D, E\}$, $R2 = \{B, F, G, H\}$, $R3 = \{D, I, J\}$
- $D3 = \{R1, R2, R3, R4, R5\}$;
 - $R1 = \{A, B, C, D\}$, $R2 = \{D, E\}$, $R3 = \{B, F\}$, $R4 = \{F, G, H\}$, $R5 = \{D, I, J\}$

DISCUSSION OF NORMALIZATION ALGORITHMS: PROBLEMS:

- We have to specify *all* the relevant functional dependencies among the database attributes.
- These algorithms are *not deterministic* in general.
- It is not always possible to find a decomposition into BCNF relation schemas that preserves dependencies.



MULTIVALUED DEPENDENCIES AND FOURTH NORMAL FORM

(A) THE EMP RELATION WITH TWO MVDs: $\text{ENAME} \twoheadrightarrow \text{PNAME}$ AND $\text{ENAME} \twoheadrightarrow \text{DNAME}$.

(B) DECOMPOSING THE EMP RELATION INTO TWO 4NF RELATIONS

(a) **EMP**

<u>ENAME</u>	PNAME	<u>DNAME</u>
--------------	-------	--------------

Smith	X	John
Smith	Y	Anna
Smith	X	Anna
Smith	Y	John

(b) **EMP_PROJECTS**

<u>ENAME</u>	PNAME
--------------	-------

Smith	X
Smith	Y

EMP_DEPENDENTS

<u>ENAME</u>	<u>DNAME</u>
--------------	--------------

Smith	John
Smith	Anna



MULTIVALUED DEPENDENCIES (MVD)

- **MVD** $X \twoheadrightarrow Y$ specifies the following constraint on any state r of R :

If two tuples t_1 and t_2 exist in r such that $t_1[X] = t_2[X]$, then two tuples t_3 and t_4 should also exist in r with the following properties, where we use Z to denote $(R - (X \cup Y))$:

- $t_3[X] = t_4[X] = t_1[X] = t_2[X]$.
- $t_3[Y] = t_1[Y]$ and $t_4[Y] = t_2[Y]$.
- $t_3[Z] = t_2[Z]$ and $t_4[Z] = t_1[Z]$.



EXAMPLE: MVD

Customer(name, addr, phones, itemsLiked)

- A customer's phones are independent of the items they like.
 - name->>phones and name->>itemsLiked.
- Thus, each of a customer's phones appears with each of the item they like in all combinations.
- This repetition is unlike FD redundancy.
 - name->addr is the only FD.



TUPLES IMPLIED BY NAME->>PHONES

If we have tuples:

name	addr	phones	itemsLiked
sue	a	p1	b1
sue	a	p2	b2
sue	a	p2	b1
sue	a	p1	b2

Then these tuples must also be in the relation.



FOURTH NORMAL FORM

- Relation R is in **4NF** with respect to a set of dependencies F if, for every *nontrivial* multivalued dependency $X \twoheadrightarrow Y$ in F^+ , X is a superkey for R .
 - F^+ includes functional dependencies and multivalued dependencies
- An MVD $X \twoheadrightarrow Y$ in R is called a **trivial MVD** if
 - a) Y is a subset of X , or
 - b) $X \cup Y = R$.
- Example:** EMP_PROJECTS has the trivial MVD $\text{Ename} \twoheadrightarrow \text{Pname}$.
- An MVD that satisfies neither (a) nor (b) is called a **nontrivial MVD**.

EMP_PROJECTS

<u>ENAME</u>	<u>PNAME</u>
Smith	X
Smith	Y

FOURTH NORMAL FORM

DECOMPOSING A RELATION STATE OF EMP THAT IS NOT IN 4NF. (A) EMP RELATION WITH ADDITIONAL TUPLES. (B) TWO CORRESPONDING 4NF RELATIONS EMP_PROJECTS AND EMP_DEPENDENTS.

(a) **EMP**

<u>ENAME</u>	PNAME	<u>DNAME</u>
--------------	-------	--------------

Smith	X	John
Smith	Y	Anna
Smith	X	Anna
Smith	Y	John
Brown	W	Jim
Brown	X	Jim
Brown	Y	Jim
Brown	Z	Jim
Brown	W	Joan
Brown	X	Joan
Brown	Y	Joan
Brown	Z	Joan
Brown	W	Bob
Brown	X	Bob
Brown	Y	Bob
Brown	Z	Bob

(b) **EMP_PROJECTS**

<u>ENAME</u>	<u>PNAME</u>
--------------	--------------

Smith	X
Smith	Y
Brown	W
Brown	X
Brown	Y
Brown	Z

EMP_DEPENDENTS

<u>ENAME</u>	<u>DNAME</u>
--------------	--------------

Smith	Anna
Smith	John
Brown	Jim
Brown	Joan
Brown	Bob

LOSSLESS (NON-ADDITIVE) JOIN DECOMPOSITION INTO 4NF RELATIONS:

The relation schemas R_1 and R_2 form a lossless (non-additive) join decomposition of R with respect to a set F of functional *and* multivalued dependencies if and only if

$$(R_1 \cap R_2) \twoheadrightarrow (R_1 - R_2)$$

or by symmetry, if and only if

$$(R_1 \cap R_2) \twoheadrightarrow (R_2 - R_1)).$$

